

MAT 534 HW02

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Problem 1. Let H and K be subgroups of a finite group G . Let $HK = \{hk : h \in H, k \in K\}$, which need not be a subgroup of G . Show that

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}.$$

Solution. Let h_1 and h_2 be in H . Then, the left cosets h_1K and h_2K are equal if and only if $h_2^{-1}h_1K = K$, which is true if and only if $h_2^{-1}h_1 \in K$, hence $h_2^{-1}h_1 \in H \cap K$. Defining a relation $\sim$ on H by decreeing $h_1 \sim h_2$ if $h_2^{-1}h_1$ tells us that the equivalence classes correspond to cosets of the subgroup $H \cap K$ in H ; in particular there are $|H|/|H \cap K|$ cosets of $H \cap K$. On the other hand each coset of $H \cap K$ corresponds to a coset of K by an element of H , according to the first sentence, and these cosets of course partition HK , so $|HK| = |K| \cdot (|H|/|H \cap K|)$ as desired. $\square$

Problem 2. Let G be a group. Define a new operation on G by

$$x * y = yx.$$

- (a) Show that G is a group under the new operation. We write G^{op} for the set G with the new operation $*$.
- (b) Show that G^{op} is isomorphic to G .
- (c) Suppose that G acts on X by a left action, and let $\alpha: G \times X \rightarrow X$ be the map which defines the action; in other words, if gx denotes the action of g on x , then $\alpha(g, x) = gx$. Show that α defines a *right* action of G^{op} on X .

Solution. (a) The operation is associative because

$$(x * y) * z = (yx) * z = zyx = x * (zy) = x * (y * z).$$

Its identity is the same as that of G because $1 * x = x \cdot 1 = x = 1 \cdot x = x * 1$, and the inverse of $g \in G$ is still g^{-1} , its inverse under $\cdot$, because $g * g^{-1} = g^{-1} \cdot g = 1 = g \cdot g^{-1} = g^{-1} * g$.

(b) Consider the map $\varphi: G \rightarrow G^{\text{op}}$ by $\varphi(g) = g^{-1}$. This is of course a bijection; it is a homomorphism because

$$\varphi(xy) = (xy)^{-1} = y^{-1}x^{-1} = \varphi(x) * \varphi(y).$$

(c) To check that α defines a right action, we need to check that $\alpha(1, x) = x$ for all x (which is immediate because α a priori defines a left action of G on X) and

$$\alpha(h, \alpha(g, x)) = \alpha(g * h, x).$$

This is easy to see because we know from α defining a left action of G on X that

$$\alpha(h, \alpha(g, x)) = \alpha(hg, x) = \alpha(g * h, x).$$

$\square$

Problem 3. Let G and H be groups, and let Φ be the set of all functions $\phi: G \rightarrow H$ with the property that $\phi(1) = 1$.

(a) Let $\alpha: G \times \Phi \rightarrow \Phi$ be defined as follows: $\alpha(g, \phi)$ is the function that takes $x \in G$ to $\phi^{-1}(g)\phi(gx) \in H$. Prove that this defines an action of G on Φ . Is it a right action or a left action?

(b) Describe the set of fixed points Φ_0 for this action.

Solution. (a) $\alpha(h, \alpha(g, \phi))$ is the function sending x to $\psi^{-1}(h)\psi(hx)$ where $\psi(x) = \phi^{-1}(g)\phi(gx)$ so it is the function

$$x \mapsto (\phi^{-1}(g)\phi(gx))^{-1}(\phi^{-1}(g)\phi(ggx)) = \phi^{-1}(gh)\phi(ggx) = \alpha(gh, x).$$

Combined with the observation $\alpha(1, \phi)$ being the map $x \mapsto \phi(1)^{-1}\phi(x) = \phi(x)$, which is ϕ , we see that α defines a left action on Φ .

(b) A function $\phi \in \Phi$ is a fixed point for the action if

$$\phi(x) = \alpha(g, \phi)(x) = \phi(g)^{-1}\phi(gx)$$

for all $x, g \in G$. By taking $x = g^{-1}$ we get $\phi(g^{-1}) = \phi(g)^{-1}$ so $\phi(x) = \phi(g^{-1})\phi(gx)$ for all $x, g \in G$. We can replace g with g^{-1} and plug in $x = gh$ for any $h \in H$ to see that ϕ satisfies

$$\phi(gh) = \phi(g)\phi(h)$$

for all $g, h \in H$; that is, ϕ must be a homomorphism $G \rightarrow H$ if it is a fixed point of the action defined by α . Conversely, it is easy to see that all homomorphisms are fixed points of the action because if ϕ is a homomorphism then

$$\alpha(g, \phi)(x) = \phi(g)^{-1}\phi(gx) = \phi(g^{-1}gx) = \phi(x)$$

for all $x \in G$. Therefore the fixed points of the action given by α are precisely the group homomorphisms $\phi: G \rightarrow H$. $\square$

Problem 4. Let p be a prime, and let G be a p -group. Suppose that G acts on a finite set X . Let $X_0 \subseteq X$ be the set of fixed points of X under this action. Show that

$$|X_0| \equiv |X| \pmod{p}.$$

Solution. Let X_0 be the set of fixed points of X and $X_1, X_2, \dots, X_r \subseteq X$ the orbits of X with more than one element. By the orbit-stabilizer theorem, each $|X_j|$ for $1 \leq j \leq r$ is a divisor of $|G|$ (because elements of the orbit are in bijection with cosets of the stabilizer in G of an element of the orbit) that is bigger than 1, hence is power of p bigger than 1, so in particular is a multiple of p . The orbits also partition X , so

$$|X| = |X_0| + \underbrace{|X_1| + \dots + |X_r|}_{0 \pmod{p}} \equiv |X_0| \pmod{p}. \quad \square$$

Problem 5. Let G be a finite group of order n , let p be a prime dividing n , and let $\mathbb{Z}/p\mathbb{Z}$ be the cyclic group of order p . Let Φ be the set of all functions $\phi: \mathbb{Z}/p\mathbb{Z} \rightarrow G$ with the property that $\phi(1) = 1$.

- (a) Show that $|\Phi| = n^{p-1}$.
- (b) The group $\mathbb{Z}/p\mathbb{Z}$ acts on the set Φ , as in Problem 3. Let Φ_0 be the set of fixed points in Φ for the action of $\mathbb{Z}/p\mathbb{Z}$. Show that the cardinality of Φ_0 is divisible by p .
- (c) Show that Φ_0 is nonempty, and hence contains at least p elements. Use this to prove *Cauchy's theorem*: if p is a prime that divides the order of a finite group G , then G contains a subgroup isomorphic to the cyclic group $\mathbb{Z}/p\mathbb{Z}$.

Solution. (a) We know that $\phi(1) = 1$; the other $p - 1$ elements of $\mathbb{Z}/p\mathbb{Z}$ each have $|G| = n$ choices for their image in G , which nets a total of n^{p-1} choices.

(b) The cardinality of the set of fixed points of the action is congruent to the cardinality of Φ mod p by Problem 4, since $\mathbb{Z}/p\mathbb{Z}$ is a p -group, and we know that $p \mid n^{p-1} = |\Phi|$.

(c) As shown in Problem 3, Φ_0 consists of the homomorphisms $\mathbb{Z}/p\mathbb{Z} \rightarrow G$. Since Φ_0 contains the trivial homomorphism $\phi(n) \equiv 1_G$ for all $n \in \mathbb{Z}/p\mathbb{Z}$, it must contain at least $p - 1$ other homomorphisms, none of which are trivial. In particular each one has trivial kernel because the only two options for the kernel of a homomorphism coming out of $\mathbb{Z}/p\mathbb{Z}$ are the trivial subgroup and $\mathbb{Z}/p\mathbb{Z}$ itself, so each nontrivial homomorphism describes a way to embed $\mathbb{Z}/p\mathbb{Z}$ as a subgroup of G , proving that G has a subgroup isomorphic to $\mathbb{Z}/p\mathbb{Z}$. $\square$

Problem 6. Let G be a group, H a subgroup, and let X be the set of left cosets of H . G acts on X by the rule $(g, aH) \mapsto gaH$, a left action.

- (a) Show that G acts transitively on X .
- (b) What is the stabilizer of the coset $aH \in X$ in G ?
- (c) Show that the homomorphism $G \rightarrow S_X$ corresponding to this action has kernel

$$\bigcap_{g \in G} gHg^{-1}.$$

- (d) Suppose that G is a finite group, and that H is a proper subgroup. Suppose that $(G : H)!$ is not divisible by $|G|$. Show that G is not simple.
- (e) Suppose that G is finite, and that $(G : H)$ is the smallest prime dividing $|G|$. Show that H is normal in G .

Solution. (a) G acts transitively on X because given cosets aH and bH for $a, b \in G$, we have

$$(ba^{-1})aH = bH,$$

implying that the two cosets lie in the same orbit.

(b) The stabilizer of aH is the subgroup aHa^{-1} . It is clear that these elements all fix aH ; conversely, if $gaH = aH$, then for each $h \in H$ there is $h' \in H$ such that

$$gah = ah' \iff g = ah'h^{-1}a \in aHa^{-1},$$

which gives the opposite containment of the stabilizer within aHa^{-1} .

(c) The kernel of this homomorphism is just the elements of G whose action on X fixes every coset, i.e. g is in the stabilizer aHa^{-1} of aH for all a so $g \in \bigcap_{a \in G} aHa^{-1}$.

(d) We prove the contrapositive. Suppose G is simple. Then G has no nontrivial normal subgroup, so the homomorphism $G \rightarrow S_X$ from part (c) has kernel 1 or G . The action of G is transitive by (a), so the homomorphism is nontrivial, hence must have kernel 1. The image of the homomorphism in S_X is then isomorphic to G , so $|G|$ divides $|S_X| = |X|!$, but $|X|$ is the number of cosets of H in G , which is precisely $(G : H)$, which means we have proven $|G|$ divides $(G : H)!$.

(e) Let X be the set of cosets of H . If H is the trivial subgroup, then it is trivially normal in G . Otherwise, $|G|$ has at least one prime factor besides $(G : H)$, which must be at least $(G : H)$. Since $(G : H)! / (G : H) = ((G : H) - 1)!$ does not have any prime factors greater than or equal to $(G : H)$, $|G|$ cannot divide $(G : H)!$.

This means that the order of the image of the homomorphism $G \rightarrow S_X$ (as in (c)), being a factor of $\gcd(|S_X|, |G|) = (G : H)$, is either 1 or $(G : H)$, due to the primality of $(G : H)$. Again, the action of G on X is transitive, so the homomorphism is nontrivial, meaning the image cannot have order 1, hence must have order $(G : H)$; therefore the kernel has order $|G| / (G : H) = |H|$. By (c) the kernel is also a subgroup of H , so this proves that the kernel is exactly H , which is enough to show that H is normal in G . $\square$

Problem 7. Let $G = GL_2(\mathbb{C})$.

(a) Give a complete set of representatives for the conjugacy classes of G .

(b) Given $A \in G$, how do you determine which element of your list A is conjugate to?

Solution. (a) Each square matrix with complex entries is similar (conjugate) to a matrix in Jordan canonical form, so the conjugacy classes are represented by the matrices in Jordan canonical form that are invertible:

$$\left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : a, b \in \mathbb{C} \setminus \{0\}, a \succ b \right\} \cup \left\{ \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix} : a \in \mathbb{C} \setminus \{0\} \right\}.$$

where $a \succ b$ means that the real part of a is greater than that of b or, if the real parts are equal, then the imaginary part of a is at least as large as that of b (i.e. order a, b as pairs of real numbers in lexicographical order) so that we don't double count matrices with the same multiset of diagonal entries (which are conjugate).

(b) Let $A \in G$. If A has two distinct eigenvalues λ_1, λ_2 with $\lambda_1 \succ \lambda_2$, then its conjugacy class is represented by

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

If $\lambda_1 = \lambda_2 = \lambda$, then either $A = \lambda I$ where I is the identity or it is conjugate to

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix};$$

it can't be conjugate to λI since the conjugacy class of λI is just $\{\lambda I\}$ because $\lambda I \in Z(G)$. $\square$